THEXRA ENTERPRISE SOLUTION PROPOSAL

13-Section Formal AI Solution Brief

Date: 07/10/2026
Status: Approved Proposal

1. Executive Summary

This enterprise solution addresses Frequent unscheduled offshore turbine downtime driven by manual, error-prone paper inspection routines and lack of digital verification.. By deploying AR — THEXRA View, the client will achieve immediate workflow optimization, enhanced spatial visual analytics, and scalable long-term digital conversion.

2. Client Challenge

Frequent unscheduled offshore turbine downtime driven by manual, error-prone paper inspection routines and lack of digital verification.

3. Project Objective

Primary Goal: Eliminate paper-based inspection procedures, reduce turbine failure rates by over 40%, and achieve 100% digitized inspection compliance across offshore assets.

Target KPIs:

- Reduce processing latency by 35%
- Improve spatial user adoption to 90%+

4. Proposed Solution

Hands-free AR inspection workflow system delivering contextual 3D step-by-step guidance and digital checklists directly to offshore technicians' smart glasses.

IMPLEMENTATION PHASING:

1. Phase 1: Discovery & Baseline Sync
2. Phase 2: Core Platform Pilot
3. Phase 3: Production Go-Live

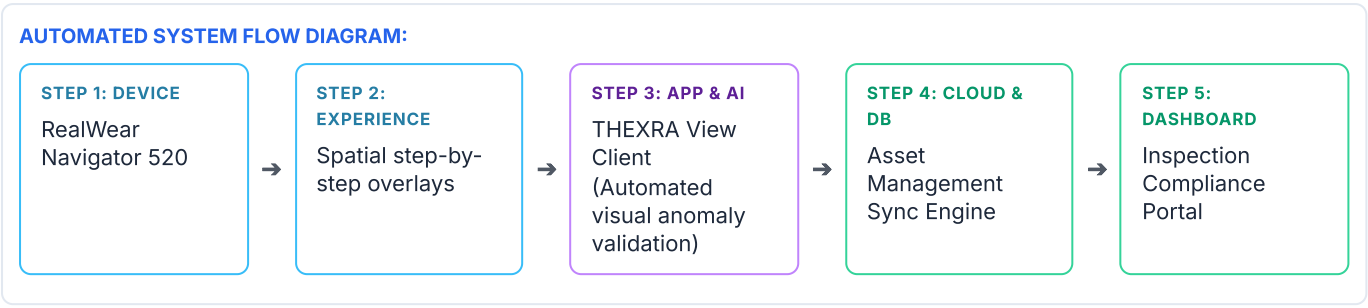
5. User Experience & Journey

Target Users: Offshore wind turbine maintenance technicians, quality control inspectors, and remote reliability engineers.

Workflow: Technician boots smart glasses at the turbine platform, selects the asset inspection route via voice command, executes hands-free AR visual checks with automated photo capture, and signs off digitally to sync audit logs.

6. Solution Architecture

AR — THEXRA View



Hardware: RealWear Navigator 520, Intrinsically Safe Industrial Tablet

Software: THEXRA View Client, THEXRA Make Content Authoring Studio

AI Components: Automated visual anomaly validation, Voice-to-text checklist transcription

XR Components: Spatial step-by-step overlays, Interactive 3D component callouts

Backend: Asset Management Sync Engine, Encrypted Offline-First Local Cache

Dashboard: Inspection Compliance Portal, Technician Work Order Management Dashboard

Data Flow: Wearable Sensor/Camera → Local Edge Runtime (THEXRA View) → Encrypted Offshore Gateway → Central Enterprise Maintenance Platform

7. Scope of Work (SOW)

Project Scope: Delivery of an end-to-end hands-free AR inspection system tailored for offshore wind turbine operations, including content authoring, hardware provisioning, and offline data sync.

Core Features: Hands-free voice-controlled checklist navigation, Visual AR step overlays on mechanical assemblies, Offline photo capture with spatial geotagging, Automated EAM work order data synchronization

Exclusions: Turbine physical hardware modifications, Satellite offshore communication infrastructure procurement, SCADA control-loop software alterations

8. Deliverables

Configured THEXRA View runtime licenses • Digitized standard operating procedure models • Admin monitoring dashboard • Technician training documentation

9. Implementation Plan

1. **Discovery** (4 weeks):Offshore workflow audit report • Technical architecture requirements
2. **Design** (4 weeks):Digital checklist UI flow • Spatial inspection asset templates
3. **Prototype** (4 weeks):Functional AR pilot module • Field validation report
4. **Development** (12 weeks):Full inspection routine build • Offline database synchronization engine
5. **Testing** (4 weeks):Offshore field trial report • Safety compliance validation signoff
6. **Deployment** (4 weeks):Fleet-wide hardware rollout • Operations training handover
7. **Support** (Ongoing):24/7 technical support SLA • Quarterly content optimization reviews

10. Timeline & Delivery Target

Target Horizon: 6-8 Weeks Deployment Target

11. Assumptions & Prerequisites

- Technical Prerequisite: API access provided by client IT team
- Operational Prerequisite: Facility network connectivity enabled

12. Optional Add-ons & Phase 2 Upsells

- Expansion Option 1: Multi-site expansion package
- Expansion Option 2: Advanced predictive AI analytics integration

13. Next Steps & Scoping Workshop

Conduct on-site offshore turbine inspection workflow mapping and technical network connectivity scoping.